PART I: FOUNDATIONS

Topic 5: Counts vs. Rates Over Time

The Core Concept

- **Count:** The raw number of events (e.g., 25 use-of-force incidents in July).
- **Rate:** The count expressed relative to an activity baseline — e.g., UoF incidents ÷ total arrests.

Why It Matters

Use-of-force incidents in Precinct North peaked in July (25). Does this mean officers became more aggressive? Not necessarily — arrests also peaked at 700 in July. More police-public contact creates more *opportunities* for force, regardless of officer behavior.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Month	UoF Count	Arrests	Rate (UoF/Arrests)	Interpretation
Jan	12	400	3.0%	Baseline
Feb	10	380	2.6%	Slightly below baseline
Mar	14	450	3.1%	Roughly stable
Apr	15	460	3.3%	Stable
May	18	550	3.3%	Stable despite more arrests
Jun	22	600	3.7%	Slight uptick
Jul	25	700	3.6%	Peak count; rate nearly stable
Sep	19	500	3.8%	Rate slightly elevated
Oct	16	420	3.8%	Rate holding
Nov	14	390	3.6%	Returning toward baseline
Dec	11	350	3.1%	Near baseline

While the raw count more than doubled from January (12) to July (25), the rate stayed nearly constant — hovering around 3.0–3.8% throughout the year.

💡 **The Insight:** The count increase was mechanically driven by higher activity volume, not a change in officer behavior. Without the rate, this conclusion is invisible.

Key Takeaway: Always ask: "Is this change real, or is it just proportional to how busy we were?" Rates answer this question; raw counts cannot.